

InternSetu — SEO & GEO Optimization

Judge-facing technical summary | Phase 1 + Phase 2 + Phase 3 completed

What was implemented

- Public, indexable opportunity pages: `/internships`, `/internships/:id`, `/jobs`, `/jobs/:id`, `/courses`.
- Dynamic **sitemap.xml** backed by active MongoDB opportunity records.
- **robots.txt**, canonical URLs, unique metadata, Open Graph and Twitter metadata.
- Crawler/social-bot pre-rendering through **backend/routes/seoCrawler.js**.
- Schema.org **JobPosting** and **BreadcrumbList** JSON-LD using real database data only.
- Semantic HTML structure for machine-readable and generative-search discovery.
- Invalid opportunity IDs return proper HTTP **404** instead of redirecting to `/`.

Performance optimization

Metric	Phase 2	Phase 3
Main app JS	621.44 kB	152.60 kB
Main app JS gzip	138.07 kB	29.91 kB
Reduction	—	75.4%
Chunk warning	>500 kB warning	0 warnings

Production improvements

- Route-level **React.lazy()** + **Suspense** for heavy/private features.
- Vendor chunk separation: **vendor-react**, **vendor-icons**, **vendor-helmet**.
- Lazy-loaded below-the-fold media with **loading="lazy"** and **decoding="async"**.
- 1-year immutable caching for `/assets/*`.
- Production security headers: **X-Content-Type-Options**, **Referrer-Policy**, **X-Frame-Options**, **Permissions-Policy**.
- MongoDB compound indexes for active opportunity listings.
- Reusable React **ErrorBoundary** and accessible **PageLoader**.

Data integrity & privacy preserved

- Candidate Matrix continues masking email and phone.
- Contact Unlock remains controlled by the existing **50** per-candidate logic.
- JWT/RBAC protection remains intact.
- Private routes and dashboard data are excluded from indexing and sitemap.
- No fabricated salary, location, dates, company data, metrics, awards, or social proof.

Verification

- `npm run build` → Exit code 0, 0 warnings
- `node backend/scripts/testBulkUnlockDuplicatePrevention.js` → 10/10 passed
- `node backend/scripts/testJobApplications.js` → PASS

- node backend/scripts/testIndustryEndToEnd.js → 41/41 passed, 0 failed
- node backend/scripts/testPhase2Seo.js → All Phase 2 verifications passed
- node backend/scripts/testPhase3Routes.js → All Phase 3 Route & Caching checks passed successfully

Judge takeaway

InternSetu is SEO/GEO optimized at the implementation level: public indexable opportunity content, structured data, crawler-ready rendering, dynamic sitemap, semantic content, and production performance optimization. Search ranking itself is an external outcome and is not guaranteed by the implementation.

Project: InternSetu | Stack: React 18 + Vite + Express/Node + MongoDB Atlas/Mongoose + JWT + Vercel

GitHub: <https://github.com/VishalChauhan85/internsheu/tree/main>